Hotel Demand Forecasting Project — Full Workflow Documentation

1. Data Loading

I began by loading the hotel bookings dataset into a pandas DataFrame.

Key actions:

- Imported the CSV file using `pd.read_csv()`
- Viewed the first rows with `.head()`
- Checked data types and structure using `.info()`
- Reviewed summary statistics using `.describe()`
- Identified missing values and potential issues

This step provided a complete understanding of the raw dataset.

2. Data Cleaning

I cleaned the dataset to ensure accuracy and consistency.

a. Removed duplicates

Used `.drop_duplicates()` to eliminate repeated rows.

b. Handled missing values

- Filled missing children values with 0
- Left agent/company missing values as they do not affect forecasting
- Ensured no critical fields were missing

c. Converted data types

- Converted arrival date columns into proper datetime format
- Ensured numerical columns were correctly typed (int/float)

d. Removed invalid records

- Removed rows with negative ADR

- Removed rows with impossible values (e.g., negative nights)

This step ensured the dataset was clean and ready for analysis.

3. Feature Engineering

I created new variables to support forecasting and visualization.

a. Total Nights

Combined weekday and weekend stays:

Code
```
total_nights = stays_in_weekend_nights + stays_in_week_nights
```

b. Arrival Date

Created a single datetime column:

Code
```
arrival_date = pd.to_datetime(arrival_date)
```

c. Revenue

Calculated using ADR × total nights:

Code
```
revenue = adr * total_nights
```

d. Daily Aggregations

Grouped the dataset by arrival date to create:

- daily demand
- daily ADR
- daily cancellations
- daily revenue

These daily metrics formed the foundation for forecasting and visualisation.

4. Preparing Data for Prophet

Prophet requires two columns:

- **ds** → date
- **y** → value to forecast

I prepared the daily demand data:

Code
```
prophet_df = daily_demand.reset_index()
prophet_df.columns = ['ds', 'y']
```

Then I initialized and trained the Prophet model:

Code
```
model = Prophet()
model.fit(prophet_df)
```

This allowed Prophet to learn the trend and seasonality patterns.

5. Forecasting & Visualisation (Step Five)

a. Created Future Dates

Generated 90 future days:

Code
```
future = model.make_future_dataframe(periods=90)
```

b. Generated Forecast

Code
```
forecast = model.predict(future)
```

c. Visualised the Forecast

Displayed the main forecast chart:

Code

```
model.plot(forecast)
```

d. Visualised Seasonality Components

Displayed weekly, yearly, and trend components:

Code

```
model.plot_components(forecast)
```

e. Visualised ADR, Cancellations, Revenue

Plotted each daily metric:

- ADR
- cancellations
- revenue

These visualisations completed Step Five and provided a full picture of hotel performance.

Summary

This project followed a complete data science workflow:

1. Loaded the dataset
2. Cleaned the data
3. Engineered features
4. Prepared Prophet input
5. Forecasted demand
6. Visualised demand, ADR, cancellations, and revenue

This document is ready to be saved as a PDF and uploaded to GitHub.